

11.3-11.4

11.3

We make a matrix with all the degenerate combinations that we could have as states

$$\begin{pmatrix} \langle 2, 0, 0 | \hat{H}_1 | 2, 0, 0 \rangle & \langle 2, 0, 0 | \hat{H}_1 | 2, 1, 0 \rangle & \langle 2, 0, 0 | \hat{H}_1 | 2, 1, 1 \rangle & \langle 2, 0, 0 | \hat{H}_1 | 2, 1, -1 \rangle \\ \langle 2, 1, 0 | \hat{H}_1 | 2, 0, 0 \rangle & \langle 2, 1, 0 | \hat{H}_1 | 2, 1, 0 \rangle & \langle 2, 1, 0 | \hat{H}_1 | 2, 1, 1 \rangle & \langle 2, 1, 0 | \hat{H}_1 | 2, 1, -1 \rangle \\ \langle 2, 1, 1 | \hat{H}_1 | 2, 0, 0 \rangle & \langle 2, 1, 1 | \hat{H}_1 | 2, 1, 0 \rangle & \langle 2, 1, 1 | \hat{H}_1 | 2, 1, 1 \rangle & \langle 2, 1, 1 | \hat{H}_1 | 2, 1, -1 \rangle \\ \langle 2, 1, -1 | \hat{H}_1 | 2, 0, 0 \rangle & \langle 2, 1, -1 | \hat{H}_1 | 2, 1, 0 \rangle & \langle 2, 1, -1 | \hat{H}_1 | 2, 1, 1 \rangle & \langle 2, 1, -1 | \hat{H}_1 | 2, 1, -1 \rangle \end{pmatrix} \quad (11)$$

We can get rid of all but two of these due to symmetry arguments: only parts with the same m will survive, parity means diagonals vanish (WHY?) and same l vanish.

11.4

By following the perturbations for the two states of the nitrogen, we see that the one aligned with the field is favored. There is an energy difference between them.

CHANGES IN THE STRONG FIELD LIMIT (HYPERFINE)

$$E_1^{(1)} = \langle 1 | \hat{H}_1 | 1 \rangle = (1, 0) \begin{pmatrix} 0 & -A \\ -A & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0 \quad (11.59a)$$

$$E_2^{(1)} = \langle 2 | \hat{H}_1 | 2 \rangle = (0, 1) \begin{pmatrix} 0 & -A \\ -A & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0 \quad (11.59b)$$

while the second-order shift is given by

$$E_1^{(2)} = \frac{|\langle 2 | \hat{H}_1 | 1 \rangle|^2}{E_1^{(0)} - E_2^{(0)}} = \frac{\left| (0, 1) \begin{pmatrix} 0 & -A \\ -A & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right|^2}{E_0 + \mu_e |\mathbf{E}| - (E_0 - \mu_e |\mathbf{E}|)} = \frac{A^2}{2\mu_e |\mathbf{E}|} \quad (11.60a)$$

$$E_2^{(2)} = \frac{|\langle 1 | \hat{H}_1 | 2 \rangle|^2}{E_2^{(0)} - E_1^{(0)}} = \frac{\left| (1, 0) \begin{pmatrix} 0 & -A \\ -A & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right|^2}{E_0 - \mu_e |\mathbf{E}| - (E_0 + \mu_e |\mathbf{E}|)} = -\frac{A^2}{2\mu_e |\mathbf{E}|} \quad (11.60b)$$